

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVPE604D - WATER POWER ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 In a typical hydropower layout, the component that converts hydraulic energy to mechanical energy is the:
 - A. Generator
 - B. Penstock
 - C. Turbine
 - D. Transformer
- The load duration curve is a graph between:
 - A. Load and power factor
 - B. Time and head
 - C. Load and duration
 - D. Load and cost
- 3 Trash racks are installed at intake structures to:
 - A. Increase water velocity
 - B. Reduce energy losses
 - C. Prevent entry of debris into the conveyance system
 - D. Reduce turbulence
- The penstock is subjected to which of the following loads?
 - A. Only atmospheric pressure
 - B. Negative pressure only
 - C. High internal pressure
 - D. No pressure
- Forebay is commonly used in:
 - A. Nuclear power plants
 - B. Thermal plants
 - C. Run-of-river hydropower plants
 - D. Diesel engines
- 6 The tailrace in a hydroelectric power station is used to:
 - A. Store water
 - B. Cool the generator
 - C. Discharge water after passing through turbine
 - D. Trap sediments

- 7 The control room in a power house is usually part of:
 A. Reservoir area
 B. Substructure
 C. Superstructure
 D. Intake system
- 8 Which turbine is most suitable for low head and large discharge?
 A. Pelton
 B. Francis
 C. Kaplan
 D. Turgo
- Draft tube should be designed for:
 A. Constant pressure flow
 B. Minimum velocity recovery
 C. Maximum kinetic energy recovery with no cavitation
 D. Maximum friction
- 10 Tidal power is a form of:
 A. Geothermal energy
 B. Wind energy
 C. Hydropower
 D. Thermal energy
- 11 Typical arrangements for upper reservoirs include:
 A. Underground tanks
 B. Elevated concrete tanks
 C. Natural hilltops or artificially created basins
 D. Submerged chambers
- Pumped storage plants help balance
 A. Temperature of the reservoir
 B. Pressure inside turbine blades
 C. Electricity supply and demand
 D. Water quality
- Q2. Solve the following.
- A) Elaborate any TWO types of Hydro Power Schemes with its advantages. 6
- B) Explain Load Duration Curves and Load Factor in detail. 6
- Q3. Solve the following.
- Write note on Surge Tank. 6
- Write a detail note on designing tunnels with hydraulic considerations. 6
- Q4. Solve Any Two of the following.
- A) Sketch the typical arrangement of a hydro power station and explain its different components. 6
- B) Explain the advantages of underground powerhouse structure. 6
- C) Explain the location of underground powerhouse with suitable diagram. 6
- Q5. Solve Any Two of the following.
- A) Explain the difference of Kaplan turbine and Pelton turbine. 6
- B) Explain the ways to avoid cavitation. 6
- C) Explain different types of tail races. 6

Q6. Solve Any Two of the following.

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| A) Enlist the components of pumped storage plants and explain them. | 6 |
| B) Write down the advantages of tidal power station. | 6 |
| C) Compare the tandem type and reversible type of pumped storage plants. | 6 |

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